SEQUENCING AGENT BEHAVIOR WITH POLICY

The examples in this section illustrate how policy can guide an agent's behavior by specifying which actions are permitted. In many real-world scenarios, however, it is not enough to control *what* an agent can do. We must also control *when it can do it*.

Agents typically execute tasks as a sequence of steps. As they progress, they gather information, produce intermediate results, and sometimes gain access to additional capabilities. A permission that is safe at one stage of a task may be unsafe at another. For example, an agent that is allowed to send email might need to summarize documents first. Allowing access to email too early could expose sensitive information from the agent's working context.

From an authorization perspective, sequencing is not a separate mechanism. It is a function of context. The current stage of a task, the presence of intermediate results, or the completion of prior steps can be expressed as contextual attributes in the authorization request. Policies can then evaluate these signals to determine whether the next action is permitted.

For example, a policy might permit sending email only after a summary has been completed:

```
permit send_email
when summary_status == "complete";
```

More complex scenarios may require multiple conditions. A financial agent might be permitted to initiate payment only after a reimbursement has been calculated and approved by a manager. In this case, sequencing combines task progression with multi-signature approval:

```
permit reimbursement_pay
when summary_status == "complete"
  && approvals.contains("manager_approved");
```